

MARTHA GAIUS KACHWEKA

mkachweka@gmail.com / mkachwek@andrew.cmu.edu | +250793092167 | Kigali,Rwanda
[linkedin.com/in/eng-martha-kachweka](https://www.linkedin.com/in/eng-martha-kachweka)

RESEARCH INTEREST

Machine learning and computer vision for healthcare; transformer-based architectures for medical image analysis; data-intensive and computational approaches to scalable diagnostic systems in low-resource environments

EDUCATION

Master of Science in Information Technology August 2023 - May 2025
Carnegie Mellon University, Pittsburgh, PA, USA

BS c. Computer Science and Engineering October 2011 - May 2016
St Joseph College of Eng. and Tech. – Dar es Salaam, Tanzania.

RESEARCH EXPERIENCE

Research Associate – Malaria Lab August 2024 - Present
Carnegie Mellon University Africa | Kigali, Rwanda

- Collected and annotated 2000+ microscopic images to build a high-quality malaria parasite detection dataset, supporting AI-driven diagnostics in low-resource settings.
- Designed and evaluated transformer-based object detection models (RT-DETR) for automated detection and classification of malaria parasites in thin and thick blood smear images.
- Conducted comparative experiments across three training strategies (baseline, online augmentation, and Distribution Focal Loss adjustments) to assess trade-offs in detection coverage and localization accuracy.
- Expanded a dataset from 2,738 to 7,419 images using structured offline augmentation, improving model robustness across diverse smear conditions.
- Achieved mAP@0.5 of 0.853 and recall of 0.844 under baseline configuration; DFL adjustments improved localization performance (mAP@50–95 = 0.589), particularly for less prevalent parasite classes.
- Performed per-class performance analysis across *P. falciparum* (PF), *P. malariae* (PM), *P. ovale* (PO), *P. vivax* (PV), and white blood cells (WBCs), demonstrating consistently high WBC detection (mAP@0.5 > 0.89).
- Preparing a research manuscript on transformer-based object detection for scalable malaria diagnostics in low-resource healthcare settings.

Collaborative Researcher- Multimodal Generative AI for Breast Cancer July 2025 - Present
CyLab Africa | Kigali, Rwanda

- Contributed to the design of a multimodal AI pipeline integrating imaging (MRI, mammography, histopathology), genomic (mRNA, CNV, mutations), clinical, and environmental data for breast cancer modeling in African populations.
- Participated in the development and evaluation of generative AI architectures (Mixture-VAE, Conditional VAE, Bayesian Flow Networks, and diffusion-based models) to generate synthetic, population-specific breast cancer datasets.
- Supported exploratory data analysis (EDA) and cross-modal data fusion strategies to address data scarcity and class imbalance in low-resource healthcare environments.
- Investigated synthetic data generation for early detection and subtype classification, with emphasis on fairness and representation in African clinical contexts.
- Collaborated with interdisciplinary teams on architecture selection, modeling strategies, and research planning for large-scale dataset curation across regional hospitals.

PUBLICATIONS

- E. Ajayi, **M. Kachweka**, M. Deku, and E. Aiken,
A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media.
AAAI Workshop on AI in Medicine and Healthcare (AIMedHealth26), January 2026.
Best Paper Award.
arXiv:2511.20001

CONFERENCE PRESENTATIONS

- Sagbo, C. P., Ndiaye, N. A., Ahabwe, A. L., **Kachweka, M.**, Olatundji, D., Shumba, N., & Gueye, A. Multimodal-Based Learning Generative AI in Breast Cancer Research for Africa.
Accepted for presentation at the Global Digital Health Forum (GDHF 2025), Nairobi, Kenya, December 3–5, 2025.

TECHNOLOGY

- Programming & Data:** Python, SQL, Pandas, NumPy, SciPy, Statsmodels, Scikit-learn.
- Machine Learning & AI:** TensorFlow, Keras, PyTorch; NLP: NLTK, TextBlob, Vader, Regex, Network Analysis: NetworkX
- Visualization & Geospatial:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Shapely, Geopandas.
- Cloud & AIOps:** Kubernetes (K8s), GCP, GKE, Istio, Prometheus, Grafana, Docker
- Productivity & Other Tools:** Microsoft 365 (Word, Excel, PowerPoint, Outlook, Teams, SharePoint), Google Workspace- (Docs, Sheets, Slides), Slack, Zoom

AWARDS AND SCHOLARSHIPS

- Masters Scholarship – CyLab-Africa / Upanzi Scholarship (2023–2025)
- Women Techsters Fellowship – Data Engineering Track (2025–2026)
- Certified Professional Engineer – Engineers Registration Board Tanzania (2021)
- Best Female Student – Structured Engineers' Apprenticeship Program (SEAP), Tanzania
- Best Undergraduate Project – Tanzania Women Engineers Convention and Exhibition (TAWECE), 2015.

TEACHING EXPERIENCE

Graduate Teaching Assistant – Introduction to Information Security

August 2024 - December 2024

CMU Africa, Kigali, Rwanda

- Supported instruction for 60+ students through grading, logistics coordination, and academic office hours.

LEADERSHIP & OUTREACH

Technology Volunteer – MasterCard Foundation Outreach Program

May 2024

Mahama Refugee Camp, Rwanda

- Delivered digital literacy training to 60+ high school students, covering Microsoft Word, Excel, PowerPoint, and introductory programming.
- Designed structured, hands-on learning sessions and supervised student capstone projects.
- Mentored students on university application pathways in science and technology fields.

PROFESSIONAL EXPERIENCE

Junior Data Engineer

February 2023 - August 2023

NMB Bank Plc, Dar es Salaam, Tanzania

- Designed and maintained SQL-based data pipelines for ETL/ELT processes.
- Conducted predictive analytics (customer churn modeling) and developed decision-support dashboards.

HQ Systems Administrator (Full Time Contract)

January 2020 - January 2023

NMB Bank Plc, Dar-es-Salaam, Tanzania.

- Managed enterprise IT infrastructure, incident response, and systems monitoring across headquarters and branch networks.

Graduate Apprentice – ICT Support (SEAP Program)

January 2018 - December 2021

NMB Bank Plc, Dar-es-Salaam, Tanzania.

- Completed the Structured Engineers' Apprenticeship Program (SEAP), integrating full-time IT systems administration with professional engineering training.
- Rotated across infrastructure, cybersecurity, and data management functions, strengthening systems-level problem-solving skills.
- Led a Windows 10 upgrade initiative for 200+ headquarters workstations, coordinating implementation under structured project timelines.
- Awarded Certified Professional Engineer status by the Engineers Registration Board of Tanzania.